

18. Construct a C program to simulate producer consumer problem using semaphores.

AIM :

To construct a C program to simulate producer consumer problem using semaphores.

PROGRAM :

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>
#define BUFFER_SIZE 5
#define MAX_ITEMS 1
int buffer[BUFFER_SIZE] = {0};
sem_t empty, full;
pthread_mutex_t mutex;
int produced_items = 0;
int consumed_items = 0;
void *producer(void *arg)
{
    while (produced_items < MAX_ITEMS)
    {
        sem_wait(&empty);
        pthread_mutex_lock(&mutex);

        for (int i = 0; i < BUFFER_SIZE; i++)
        {
            if (buffer[i] == 0)
            {
                buffer[i] = produced_items + 1;
```

```

        printf("Produced: %d\n", buffer[i]);
        produced_items++;
        break;
    }
}

pthread_mutex_unlock(&mutex);
sem_post(&full);
sleep(1);
}
return NULL;
}

void *consumer(void *arg)
{
    while (consumed_items < MAX_ITEMS)
    {
        sem_wait(&full);
        pthread_mutex_lock(&mutex);

        for (int i = 0; i < BUFFER_SIZE; i++)
        {
            if (buffer[i] != 0)
            {
                printf("Consumed: %d\n", buffer[i]);
                buffer[i] = 0;
                consumed_items++;
                break;
            }
        }

        pthread_mutex_unlock(&mutex);
        sem_post(&empty);
    }
}

```

```
        sleep(2);
    }
    return NULL;
}

int main()
{
    pthread_t producer_thread, consumer_thread;
    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&mutex, NULL);
    pthread_create(&producer_thread, NULL, producer, NULL);
    pthread_create(&consumer_thread, NULL, consumer, NULL);
    pthread_join(producer_thread, NULL);
    pthread_join(consumer_thread, NULL);
    pthread_mutex_destroy(&mutex);
    sem_destroy(&empty);
    sem_destroy(&full);
    return 0;
}
```

OUTPUT:

Enter Input here

If your code takes input, add it in the above box before running.

Output

Status : Successfully executed

Time:

0.0000 secs

Memory:

1.984 Mb

Your Output

Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Produced: 4
Consumed: 3
Produced: 5
Produced: 6
Consumed: 5

If your code takes input, add it in the above box before running.

Output

Status : Successfully executed

Time:

0.0000 secs

Memory:

1.984 Mb

Your Output

Produced: 7
Produced: 8
Consumed: 7
Produced: 9
Produced: 10
Consumed: 9
Consumed: 4
Consumed: 6
Consumed: 8
Consumed: 10